# Algebraic Topology

Shing Tak Lam

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Lecture 1

## 1 Introduction

Let $X, Y$ be topological spaces, $f, g : X \to Y$ be continuous functions. Note we will use the terminology 'map := continuous function'. We say that $f$ *is homotopic to* $g$, written $f \simeq g$, if there exists a map $F : X \times I \to Y$, such that

$$F|_{X\times\{0\}} = f \quad \textit{and} \quad F|_{X\times\{1\}} = g$$

Here $I = [0, 1]$ carries its Euclidean topology. An exercise (sheet 1) is that this defines an equivalence relation on the set of maps $X \to Y$.

**Definition 1.1** (homotopy equivalence)
We say that $f : X \to Y$ is a *homotopy equivalence* if there exists a function $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We say that $g$ is a *homotopy inverse* to $f$.

The same exercise on sheet 1 shows that homotopy equivalence is an equivalence relation on topological spaces.

**Example 1.2**
1. If $f : X \to Y$ is a homeomorphism, then $f$ is also a homotopy equivalence.
2. $i : \{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivance.
3. The inclusion $i : S^{n-1} \hookrightarrow \mathbb{R}^n \setminus 0$ is a homotopy equivalence.

**Definition 1.3**
A space $X$ is *contractible* if it is homotopy equivalent to a point.

Idea of Algebraic Topology: Study spaces up to homotopy equivalence. Broadly, we are interested in "connectivity properties" of topological spaces.

**Example 1.4**
1. We say a space $X$ is *path-connected* if any two maps $\{*\} \to X$ are homotopic. For example, $\mathbb{R}$ is path connected, $\mathbb{R} \setminus \{0\}$ is not. A corollary of this is the intermediate value theorem.
2. We say a path connected space $X$ is *simply connected* if every map $f : S^1 \to X$ is homotopic to

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a constant map. Equivalently, every two maps $S^1 \to X$ are homotopic. Or equivalently (again) every continuous map $S^1 \to X$ extends to a continuous map $D^2 \to X$. For example, $\mathbb{R}^2$ is simply connected, by $\mathbb{R}^2 \setminus \{0\}$ is not. If $\gamma : S^1 \to \mathbb{R}^2 \setminus \{0\}$ is continuous, it has a *winding number*, or *degree* $\deg(\gamma) \in \mathbb{Z}$, such that

- It is invariant under homotopy,
- If $\gamma_n(t) = \exp(2n\pi it)$, then $\deg(\gamma_n) = n$

In particular, taking $n = 0$ gives the constant map. Therefore, a constant map has degree 0. A corollary of this is the fundamental theorem of algebra.

3. We say a path connected space $X$ is *$k$-connected*, if for all $i \leq k$, every map $S^i \to X$ is homotopic to a constant map. In this case, $\mathbb{R}^n$ is $(n-1)$-connected, but $\mathbb{R}^n \setminus 0$ is not. More precisely, maps from $S^{n-1} \to \mathbb{R}^n \setminus 0$ have a degree $\deg(f) \in Z$, such that degree is a homotopy invariant, and the degree of the inclusion map is 1, and the degree of the constant map is 0. A corollary of this is the Brouwer fixed point theorem.

Lecture 2

## 1.1 Co(chain) complexes

We will define invariants of topological spaces in two steps:

(a) We associate to $X$ a (co)chain complex,

(b) We take the (co)homology of the complex

The topology enters in step (a), whereas (b) is just homological algebra.

**Definition 1.5** (chain complex)

A *chain complex*@ $(C_*, \partial)$ consists of a sequence $\{C_i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial_n : C_n \to C_{n-1}$ for all $n$, satisfying $\partial_n \circ \partial_{n+1} = 0$. We also write this conditions as $\partial^2 = 0$.

**Definition 1.6** (homology group)

The *$i$-th homology group* of a chain complex $(C_*, \partial)$ is

$$\mathrm{H}_i(C_*, \partial) = \frac{\ker(\partial_i : C_i \to C_{i-1})}{\mathrm{im}(\partial_{i+1} : C_{i+1} \to C_i)}$$

We write

$$\mathrm{H}_*(C_*, \partial) = \bigoplus_{i\in\mathbb{Z}} \mathrm{H}_i(C_*, \partial)$$

**Definition 1.7** (cochain complex)

A *cochain complex* $(C^*, \partial)$ consists of a sequence $\{C^i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial^n : C^n \to C^{n+1}$ for all $n$, satisfying $\partial^{n+1} \circ \partial^n = 0$. We also write this conditions as $\partial^2 = 0$.

**Definition 1.8** (cohomology group)

The *$i$-th cohomology group* of a cochain complex $(C^*, \partial)$ is

$$\mathrm{H}^i(C^*, \partial) = \frac{\ker(\partial^i : C^i \to C^{i+1})}{\mathrm{im}(\partial^{i-1} : C^{i-1} \to C^i)}$$

We write

$$H^*(C^*, \partial) = \bigoplus_{i \in \mathbb{Z}} H_i(C^*, \partial)$$

**Notation 1.9.** Elements of $\ker(\partial)$ are called *cycles* (in a chain complex), *cocycles* (in a cochain complex). Elements of $\mathrm{im}(\partial)$ are called *boundaries* (and *coboundary* resp.).

Elements of $H_*$ are called *homology classes*, and elements of $H^*$ are called *cohomology classes*.

We will call $\partial$ the *differential*.

**Definition 1.10** (chain map)

Given chain complex $(C_*, \partial)$, $(D_*, \partial)$, a *chain map* $f : C_* \to D_*$ comprises group homomorphism $f_i : C_i \to D_i$, such that

$$\begin{array}{ccc} C_i & \xrightarrow{f_i} & D_i \\ \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} \\ C_{i-1} & \xrightarrow[f_{i-1}]{} & D_{i-1} \end{array}$$

commutes.

**Lemma 1.11.** A chain map $f : C_* \to D_*$ induces homomorphisms $f_* : H_i(C_*) \to H_i(D_*)$.

*Proof.* Let $a \in H_i(C_*)$, and we can choose a cycle $\alpha \in C_i(C_*)$, with $\partial\alpha = 0$. In this case,

$$\partial f_i(\alpha) = f_{i-1}(\partial\alpha) = f_{i-1}(0) = 0$$

So $f_i(\alpha) \in D_i$ is a cycle. We set $f_*(a) = [f_i(\alpha)] \in H_i(D_*)$. Next, we need to show that this is independent of choices. Suppose $a = [\alpha] = [\alpha']$. Then $\alpha - \alpha' \in \mathrm{im}(\partial)$. Therefore, we can write $\alpha - \alpha' = \partial\tau$. In this case,

$$f_i(\alpha) - f_i(\alpha') = f_i(\partial\tau) = \partial f_{i+1}(\tau) \in \mathrm{im}(\partial)$$

So we conclude $[f_i(\alpha)] = [f_i(\alpha')] \in H_i(D_*)$. It is clear that $f_*$ defines a homomorphism. □

Correspondingly, we can define *cochain maps* $C^* \to D^*$, that is, $f^i : C^i \to D^i$ such that

$$\begin{array}{ccc} C^{i+1} & \xrightarrow{f^{i+1}} & D^{i+1} \\ \uparrow{\scriptstyle \partial} & & \uparrow{\scriptstyle \partial} \\ C^i & \xrightarrow[f^i]{} & D^i \end{array}$$

commutes.

Exercise: This construction is functorial. That is, $(\mathrm{id}_{C_*})_* = \mathrm{id}_{H_*}$, and if we have $f : C_* \to D_*$, $g : D_* \to E_*$, then $(g \circ f)_* = g_* \circ f_*$.

Goal: To associate to a topological space $X$ (co)chain complex $C_*(X)$ and $C^*(X)$. The definition works for any $X$, but the theories are better behaved for "nicer" $X$.

## 1.2 Singular (co)homology

We will develop singular (co)homology.

**Definition 1.12** (standard simplex, face)

The *standard simplex* is

$$\Delta^n = \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid t_i \geq 0, \sum_i t_i = 1 \right\}$$

The $i$-th face $\Delta^n$ is

$$\Delta^n_i = \Delta^n \cap \{t_i = 0\}$$

and we have a canonical homeomorphism

$$\begin{aligned} \delta_i : \Delta^{n-1} &\to \Delta^{n-1}_i \\ (t_0, \dots, t_{n-1}) &\mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}) \end{aligned}$$

**Definition 1.13** (singular simplex, singular chain complex)

If $X$ is a topological space, a *singular $n$-simplex* in $X$ is a continuous map $\sigma : \Delta^n \to X$.

The *singular chain complex* is given by the the free abelian groups

$$C_i(X) = \bigoplus_{\sigma \text{ singular } n\text{-simplex}} \mathbb{Z}\sigma$$

We define the boundary map $C_i(X) \to C_{i-1}(X)$ via

$$\partial\sigma = \sum_{j=0}^{i} (-1)^j \sigma|_{\Delta^i_j}$$

where

$$\sigma|_{\Delta^i_j} := \sigma \circ \delta_j : \Delta^{n-1} \to X$$

Note if $\{v_i\}_{i=0}^n$ are $n+1$ ordered points in $\mathbb{R}^{n+1}$, and if $\{v_i - v_0\}_{i=1}^n$ are linearly independent, then the convex hull of $v_0, \dots, v_n$ is an $n$-simplex, which we will call

$$[v_0, \dots, v_n]$$

which is given by the map

$$\begin{aligned} \Delta^n &\to \mathbb{R}^{n+1} \\ t &\mapsto \sum_i t_i v_i \end{aligned}$$

We orient the edges of the standard simplex (and thus any simplex $[v_0, \dots, v_n]$) by saying that $v_i < v_j$ if $i < j$.

More concretely, for $n = 2$, we have the oriented edges $v_0 \to v_1, v_0 \to v_2, v_1 \to v_2$.

**Lemma 1.14.** $\partial^2 = 0$.

*Proof.* If $\sigma$ is an $n$-simplex, defined on $[v_0, \dots, v_n]$, then

$$\partial\sigma = \sum_j (-1)^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, v_n]}$$

and so

$$\partial(\partial\sigma) = \sum_{i<j} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, \widehat{v_j}, \dots, v_n]} + \sum_{i>j} (-1)^{i+1} - 1^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, \widehat{v_i}, \dots, v_n]} = 0$$

since the signs cancel. □

**Definition 1.15**

The *singular homology groups of* $X$, $\mathrm{H}_i(X)$, are the homology groups of the singular chain complex.

**Remark 1.16.** Since all we used is the topology on $X$, $\mathrm{H}_i(X)$ is clearly a topological (read homeomorphism) invariant of $X$.

Moreover, if $f : X \to Y$ is continuous, $\sigma : \Delta^n \to X$ is an $n$-simplex on $X$, then $f \circ \sigma$ is an $n$-simplex on $Y$. The fact that

$$(f \circ \sigma) \circ \delta_j = f \circ (\sigma \circ \delta_j)$$

shows that $f$ induces a chain map.

Lecture 3

**Definition 1.17** (singular cochain complex)

The *singular cochain complex* of $X$ has

$$C^i(X) = \mathrm{Hom}_{\mathbb{Z}}(C_i(X), \mathbb{Z})$$

and the boundary operator

$$\partial^* \psi(\sigma) = \psi(\partial \sigma)$$

for $\psi \in C^i(X)$, $\sigma \in C_{i+1}(X)$. That is, it is adjoint to $\partial : C_i(X) \to C_{i-1}(X)$.

Since $\partial^2 = 0, (\partial^*)^2 = 0$. The *singular cohomology* is

$$\mathrm{H}^*(X) = \bigoplus_{i \geq 0} \mathrm{H}^i(C^*, \partial^*)$$

If $f : X \to Y$ continuous, then $f$ induces a chain map $f_* : C_*(X) \to C_*(Y)$ via $f_*(\sigma) = f \circ \sigma$, satisfying $\partial f_* = f_* \partial$. Similarly, $f : X \to Y$ defines a *pullback map* $f^* : C^*(Y) \to C^*(X)$, given by

$$f^*(\psi)(\tau) = \psi(f_*(\tau))$$

Again, we have that

$$\partial^* f^*(\psi)(\tau) = f^* \psi(\partial \tau) = \psi(f_* \partial \tau) = \psi(\partial f_* \tau) = (\partial^* \psi)(f_* \tau) = f^*(\partial^* \psi)(\tau)$$

So $f^*$ is a cochain map, i.e. $\partial^* f^* = f^* \partial^*$. Therefore, we have an induced homomorphism on cohomology,

$$f^* : \mathrm{H}^*(Y) \to \mathrm{H}^*(X)$$

Warning: by definition,

$$C^i(X) = \mathrm{Hom}_{\mathbb{Z}}(C_i(X), \mathbb{Z})$$

but it is not true (in general) that

$$\mathrm{H}^i(X) = \mathrm{Hom}_{\mathbb{Z}}(\mathrm{H}_i(X), \mathbb{Z})$$

### 1.2.1 Basic computations

**Lemma 1.18** (homology of a point)**.** Let $X = \{\bullet\}$. Then

$$\mathrm{H}_n(X) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

*Proof.* By definition, $C_i(X)$ is the free abelian groups on continuous map $\Delta^i \to \{\bullet\}$. But there is only one such map, namely the constant map (for $i \geq 0$). Hence the chain complex looks like

$$\cdots \longrightarrow C_2(X) \longrightarrow C_1(X) \longrightarrow C_0(X) \longrightarrow 0$$

where each $C_i(X)$ is $\mathbb{Z}$. Computing the boundary maps,

$$\partial(\sigma_n) = \begin{cases} \sigma_{n-1} & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

Hence the chain complex is

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} 0$$

We can then compute the homology groups explicity. □

**Lemma 1.19.** If we write

$$X = \bigsqcup_{\alpha \in A} X_\alpha$$

as a disjoint union of path components, then

$$\mathrm{H}_i(X) = \bigoplus_{\alpha \in A} \mathrm{H}_i(X_\alpha)$$

Moreover, if $X$ is path connected (and nonempty), then $\mathrm{H}_0(X) = \mathbb{Z}$.

*Proof.* Since $\Delta^i$ is path connected, the image of a continuous map $\sigma : \Delta^i \to X$ must lie within some $X_\alpha$. Moreover, so does the image of all of the boundary faces $\sigma \circ \delta_j$. So in fact

$$(C_i(X), \partial) = \left( \bigoplus_{\alpha \in A} C_i(X_\alpha), \partial \right)$$

Note that a general element of $C_i(X)$ is a finite linear combination of $i$-simplicies in $X$. So its image can only meet finitely many of the $X_\alpha$.

Now suppose $X$ is path connected. Define a function

$$\begin{aligned} \varepsilon : C_0(X) &\to \mathbb{Z} \\ \sum n_i \sigma_i &\mapsto \sum n_i \end{aligned}$$

Since $X$ is nonempty, $\varepsilon$ is surjective. On the other hand, if $\tau \in C_1(X)$, then $\varepsilon(\tau(1) - \tau(0)) = 0$. So by linearity,

$$\mathrm{im}(\partial : C_1(X) \to C_0(X)) \subseteq \ker(\varepsilon)$$

Recall

$$\mathrm{H}_0(X) = \frac{C_0(X)}{\mathrm{im}(\partial : C_1(X) \to C_0(X))}$$

and so $\varepsilon$ descends to a map $\mathrm{H}_0(X) \to \mathbb{Z}$. Now we will use the fact that $X$ is path connected. Suppose

$$\sum n_i \sigma_i \in \ker(\varepsilon)$$

and fix a base point $p \in X$. Since the $\sigma_i$ are 0-simplicies, they correspond to points in $X$. Moreover, since $X$ is path connected, we can choose paths

$$\tau_i : \Delta^1 \to X$$

such that $\tau_i(1) = \sigma_i$ and $\tau_i(0) = p$. In this case,

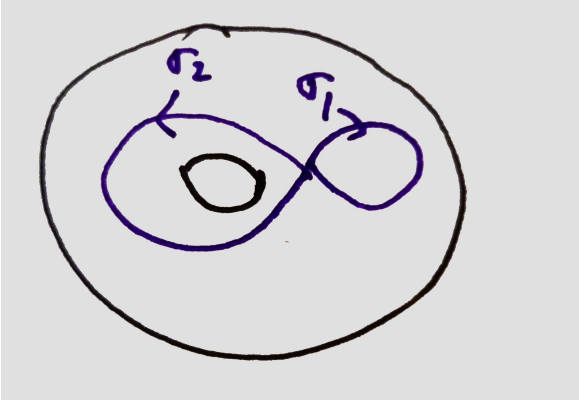
$$\partial \left( \sum n_i \tau_i \right) = \sum n_i \partial \tau_i = \sum n_i \sigma_i - (\sum n_i) p$$

But we assumed $\sum n_i = 0$, and so $\ker(\varepsilon) \subseteq \mathrm{im}(\partial : C_1(X) \to C_0(X))$. This then gives the required result, as the induced map

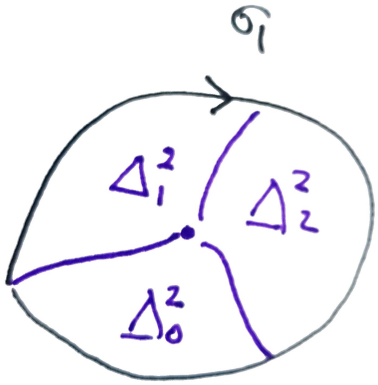
$$\mathrm{H}_0(X) \to \mathbb{Z}$$

is then an isomorphism. □

Informal conjecture: We can't compute anything else.
Therefore, we need to develop some more structure about (co)homology.
What are we doing? Consider the case of $X$ being an annulus in $\mathbb{R}^2$. We have 1-simplices




($\sigma_1$ null-homotopic, $\sigma_2$ winds around the origin)
In this case, $\partial\sigma_1 = \partial\sigma_2 = 0$, and so they define homology classes.




Moreover, $[\partial\sigma_1] = 0$, since we have triangles $\tau_0, \tau_1, \tau_2$ such that

$$\partial(\tau_0 + \tau_1 + \tau_2) = \sigma_1$$

From this intuition, $[\sigma_2] \neq 0$. But to show even this, we need some structure theorems.

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